

Bokwidi Old Age Centre (BOAC)

Project progress report - website, admin desk and volunteer desk

Module	INSY7315 - 2026 Task 2, Group 4
Report date	26 September 2026
Branch / commit	feature-backend @ 268895d (working tree clean)
Period covered	23 September - 26 September 2026 (34 commits)
Contributors	Unathi Tshuma (ST10377293), Jordan Harding (ST10439633), Bokamoso Sebake
Sources	docs/work-log (17 notes), docs/research (2 notes), git history, current source tree

This report brings together every work-log and research note in **docs/** with the git history. The work done on 26 September (public-page wiring, request validation and the frontend folder rename) has no work-log note, so section 7 rebuilds it from the commits and checks that nothing was lost in the rename.

1. Status at a glance

Area	State	Evidence
Hosting	Live on Railway (project just-truth). API deploys from feature-backend, root /backend.	Deploy ccc22934 SUCCESS; /health returned {database:true, bucket:true}
Database	MongoDB 8.0 on Railway, database boac, strict JSON-Schema validators on every collection.	npm run db:setup printed 'Database schema is ready'
File storage	S3-compatible bucket boac-uploads (EU West) for content covers and application images.	HeadBucket health check passed
Auth	Better Auth with email/password, public sign-up off, roles admin / editor / reviewer / volunteer.	Sign-in, 401 and 403 checks run against localhost
Backend API	Hono on Node. Content, taxonomy, opportunities, applications, assignments, volunteer work, contact messages. Zod validation on all inputs.	tsc --noEmit exit 0 (re-run 26 Sep)
Frontend	Vite + React 19 app: public pages, admin desk and volunteer desk behind one sign-in. TanStack Query and a typed Hono client.	tsc --noEmit exit 0 (re-run 26 Sep)
Docs	17 work-log notes up to 25 Sep; 2 research notes. 26 Sep work not yet written up.	See section 7

2. Timeline from the commits

Date	Commit(s)	Author	Work
23 Sep	4b99d27 - 3983e76	B. Sebake	README, Task 1.3 (IA and shared CMS content model) and Task 2.1 (responsive wireframes) for BOAC; main then reverted to blank, work moved to feature/bk-uiux-design.
24 Sep	b77ce88, 3391b79	U. Tshuma	Documentation; Hono backend starter.
24 Sep	ef72fe3, 5dd56ce	U. Tshuma	MongoDB and bucket clients; research into CMS/volunteer models and the MongoDB schema.
24 Sep	6fbaf7d, f88d6b1, 265d299	U. Tshuma	Listen on Railway PORT; hosting docs; linking the api service to MongoDB and the bucket.
24 Sep	f71d5c9, 837e87e, 8ca4233, 4b16d9c	U. Tshuma	Admin authentication, then replaced by Better Auth for the React client; docs.
24 Sep	a19ca21	U. Tshuma	Volunteer sign-in and personal work tracking.
25 Sep	b685f05 - b382a0c	U. Tshuma	Admin/CMS routes, application management, RBAC, content routes, opportunities, schema fields, bucket storage for images.
25 Sep	ff0b4d1	U. Tshuma	Volunteer dashboard API and the React admin desk (frontend created).
25 Sep	1ae03e3, a75f998	U. Tshuma	Volunteer schedule / work / spaces pages; one shared sign-in.
25 Sep	5493eb7, beee8a9, 5623189, 9afe92c	J. Harding	Get Involved, Volunteer Form, Donation and Contact pages.
26 Sep	e5b2df4	U. Tshuma	Merge feature-frontend into feature-backend.
26 Sep	5a6a6af	U. Tshuma	Public pages connected to the backend; contact messages; build fixes.
26 Sep	28d3664	U. Tshuma	Zod validation and uniform { error } responses across the API.
26 Sep	268895d	U. Tshuma	Frontend restructured into area folders with kebab-case names.

3. Infrastructure and deployment

All hosting is on Railway, project **just-truth**, environment **production**, region EU West.

Resource	Detail
BOAC service (API)	https://boac-production.up.railway.app. GitHub trigger on zerobreak/BOAC branch feature-backend; root /backend; build npm run build; start npm start (Railpack). Listens on process.env.PORT or 3000.
MongoDB	mongo:8.0 service with volume mongodb-volume. Public TCP proxy iriguchi.proxy.rlwy.net:24835 for local development.
Bucket boac-uploads	Region ams, S3 endpoint https://t3.storageapi.dev, checksums WHEN_REQUIRED.
Service links	The api service reads MONGO_URL and the AWS_* bucket variables as Railway reference variables, which draws the connections on the canvas.
Git remotes	origin pushes to both EMGPMD/insy7315-2026-task-2-group-4 and zerobreak/BOAC.

Secrets (MONGODB_URI, BETTER_AUTH_SECRET, admin and volunteer passwords, bucket keys) live only in backend/.env and on Railway. They are not in the repo or in this report.

4. Backend

Stack: Hono 4 on @hono/node-server, TypeScript, MongoDB driver, @aws-sdk/client-s3, Better Auth with the Mongo adapter and admin plugin, Zod with @hono/zod-validator.

4.1 Data model (database boac)

Collection	Purpose and key rules
roles	admin, editor, reviewer, volunteer with embedded permission lists (manage_content, publish_content, manage_opportunities, review_applications, track_work).
user / account / session	Better Auth collections. Passwords hashed in the credential account.
content	type post article, status draft published, authorId, optional categoryId, up to 20 tagIds, optional coverKey.
categories, tags	Unique names; referenced by id from content.
opportunities	title, location, status open closed, public flag, closingDate. No delete route.
volunteer_applications	Linked to an opportunity. Status submitted, under_review, needs_info, declined, accepted. Up to 3 private image fileKeys; skills, availability, motivation; internalNotes. Partial unique index on email + opportunityId unless declined.
volunteer_work	Owned by volunteerId. status planned in_progress done, notes, opportunityId, startsAt, hours.
contact_messages	name, email, phone, subject (general, volunteer, donation, partnership), message, status new handled.

4.2 Authentication and roles

- Better Auth mounted at /api/auth; sign-in is POST /api/auth/sign-in/email; public sign-up is disabled.
- CORS allows FRONTEND_ORIGIN (http://localhost:5173 by default) with credentials.
- requireAdmin and requireVolunteer return 401 with no session and 403 for the wrong role. requireStaff checks the Better Auth permission (manage_content, publish_content).
- admin@boac.local and volunteer@boac.local are seeded from env on setup and on server start. Accepting an application provisions a volunteer account and returns a one-time password once.

4.3 API routes

Area	Public	Signed-in
Content	GET /content, /content/:id (published only), /categories, /tags, /content/:id/cover	admin: CRUD /admin/content, cover upload/delete, /admin/categories, /admin/tags
Opportunities	GET /opportunities (public and open only)	admin: list, create, patch

Area	Public	Signed-in
Applications	POST /applications (multipart, up to 3 images, 5 MB each)	admin: list, get, status patch, file download; accept creates volunteer + work item
Assignments	-	admin: POST /admin/assignments (volunteer, opportunity, optional startsAt)
Contact	POST /contact	admin: GET /admin/messages, PATCH status
Volunteer	-	volunteer: GET /volunteer/dashboard, GET/POST /volunteer/work, PATCH /volunteer/work/:id
Health	GET /health (database and bucket)	-

4.4 Validation and errors (26 Sep)

- backend/src/validate.ts centralises validate(), objectId(), idParam, required(), email and optionalDate().
- Every failed check answers 400 with { error: '<first message>' }; a non-ObjectId route id answers 404.
- app.onError turns malformed bodies and unexpected failures into the same { error } shape (500 is logged).
- The full route chain is exported as AppType so the frontend client is typed from it.

5. Frontend

Vite + React 19 + React Router 7 app on port 5173, using the better-auth React client, TanStack Query and hono/client. Because the client imports AppType from the backend, renaming a route or field breaks the frontend build instead of failing at runtime.

5.1 Routes

Who	Pages
Anyone	/get-involved, /volunteer-form (posts to /applications with the chosen opportunity), /donate, /contact (posts to /contact)
Sign-in	/login - one form for every account; the role on the account decides which desk opens.
Admin	Desk, Content, Library (categories and tags), Opportunities, Applications, Assign, Messages
Volunteer	Schedule, Work, Spaces

5.2 Folder layout after the rename (268895d)

Folder	Holds
src/app/	app.tsx (route table), shell.tsx (signed-in shell), public-layout.tsx, styles.css
src/lib/	api-client.ts, query-client.ts, auth.ts, access.ts
src/admin/	admin pages, admin-routes.tsx, queries.ts
src/volunteer/	volunteer pages, volunteer-routes.tsx, queries.ts, format.ts
src/public/	public pages, queries.ts
src/auth/	login-page.tsx

File names are kebab-case; components stay PascalCase. Each page sits in the folder of its area with that area's queries beside it. The backend stays flat in backend/src.

6. Research that shaped the design

Public site and admin models (24 Sep). From WordPress, Drupal, Payload CMS, CiviCRM and CiviVolunteer documentation:

- Authorise by permissions attached to roles, not by checking role names - hence the permission lists on roles and Better Auth access control.
- One content collection with a type and a draft/published status; the public only reads published items.
- Applications must reference a specific opportunity (the original ERD had no link); internal notes, reviewer and applicant details stay admin-only.
- Fixed original ERD issues: users-to-posts cardinality was reversed, articles had no status, tags was a single column.

Volunteer workspaces (25 Sep). From CiviVolunteer, Golden, VolunteerHub, Better Impact, Volunteers for Salesforce and SignUpGenius:

- Volunteers see a personal schedule of their own upcoming and finished assignments, not a roster.
- A shift time (startsAt) is set by the admin; location stays on the opportunity and becomes the 'space'.
- Volunteers record hours on their own work rows. Check-in was left out because the products disagree.
- The profile shown beside the schedule is the session user.

7. Work done on 26 September (from commits, not yet in work-log)

5a6a6af - Connect the public pages to the backend and fix the frontend build

- Get Involved, Volunteer, Donate and Contact moved outside the login.
- Volunteer form sends POST /applications with the chosen opportunity; skills, availability and motivation are kept on the application (schema fields added).
- New backend/src/contact.ts and contact_messages collection; admin Messages page.
- Dead links, missing images and the copy button fixed.

28d3664 - API error handling and validation

- Added zod and @hono/zod-validator; new backend/src/validate.ts.
- applications, assignments, contact, content, opportunities and volunteer-work rewritten to use the validators (655 insertions, 736 deletions).
- Global app.onError returns structured { error } responses.

268895d - Frontend restructure and rename

- App.tsx split into app/app.tsx, app/shell.tsx, app/public-layout.tsx, admin/admin-routes.tsx and volunteer/volunteer-routes.tsx.
- api.ts replaced by lib/api-client.ts (typed hono client) and lib/query-client.ts; pages moved to TanStack Query with per-area queries.ts.
- Added @tanstack/react-query and hono to the frontend; new frontend/README.md.

7.1 Was anything lost in the rename?

No. git show -M on 268895d maps every file from the old flat layout to a new path:

Old path (frontend/src/)	New path (frontend/src/)	Similarity
pages/ApplicationsPage.tsx	admin/applications-page.tsx	47%
pages/AssignPage.tsx	admin/assign-page.tsx	45%
pages/ContentPage.tsx	admin/content-page.tsx	44%
pages/DeskPage.tsx	admin/desk-page.tsx (rewritten)	new file
pages/TaxonomyPage.tsx	admin/library-page.tsx	33%
pages/MessagesPage.tsx	admin/messages-page.tsx	37%
pages/OpportunitiesPage.tsx	admin/opportunities-page.tsx	38%
pages/LoginPage.tsx	auth/login-page.tsx	97%
pages/ContactPage.tsx	public/contact-page.tsx	78%
pages/DonationPage.tsx	public/donate-page.tsx	98%
pages/GetInvolvedPage.tsx	public/get-involved-page.tsx	100%
pages/VolunteerFormPage.tsx	public/volunteer-form-page.tsx	71%
pages/VolunteerHomePage.tsx	volunteer/schedule-page.tsx	62%
pages/VolunteerWorkPage.tsx	volunteer/work-page.tsx	53%
pages/VolunteerSpacesPage.tsx	volunteer/spaces-page.tsx	38%
pages/volunteer.ts	volunteer/format.ts (+ queries.ts)	40%
api.ts / auth.ts / access.ts	lib/api-client.ts / lib/auth.ts / lib/access.ts	44% / 100% / 100%
styles.css	app/styles.css	100%
App.tsx	app/app.tsx, app/shell.tsx, admin-routes.tsx, volunteer-routes.tsx	split

The commit message says pages were 'deleted', but they were moved and rewritten, not removed. Both apps still pass `tsc --noEmit` after the change, and the working tree is clean, so no uncommitted work is at risk.

8. What has been verified

Check	Result	When
Railway deploy of feature-backend	SUCCESS, GET / returned Hello Hono!	24 Sep
Live /health	{ database: true, bucket: true }	24 Sep
Schema validators	Incomplete content insert rejected	24 Sep
Better Auth sign-in	Wrong password 401; admin 200 with HttpOnly cookie; sign-up blocked 400	24 Sep
Role gates	/admin 401 without cookie; volunteer route 403 for admin	24 Sep
Volunteer work ownership	Another volunteer's row hidden from the list	24 Sep
Admin / volunteer UI	Login page renders; /work with no session redirects to /login	25 Sep
Typecheck backend + frontend	Both exit 0	26 Sep (for this report)

9. Gaps and next steps

- Content, opportunity, application, assignment and dashboard routes were only typechecked when written (backend/.env was missing then). backend/.env is present again, so an end-to-end HTTP pass is the next check.
- No signed-in pass of the admin or volunteer desks has been run in a browser.
- Write work-log notes for the 26 Sep commits (public pages and contact, validation, frontend restructure).
- Push feature-backend so Railway deploys the new routes; the frontend has no hosting yet and needs VITE_API_URL and a matching FRONTEND_ORIGIN.
- Email notifications for application outcomes were left out of scope.
- Earlier work-log notes still reference the old frontend paths (for example src/pages/LoginPage.tsx); the table in 7.1 gives the current locations.